Fundamentals of Electrical and Electronics Engineering (FEEE)

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T-P-C: 3-2-4

Lecture-29

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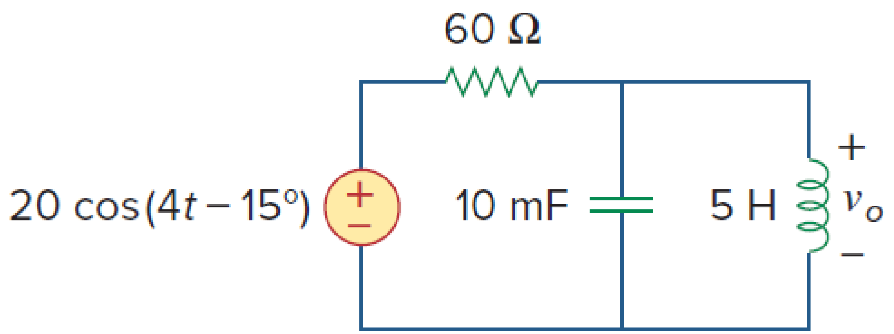
Practice Questions

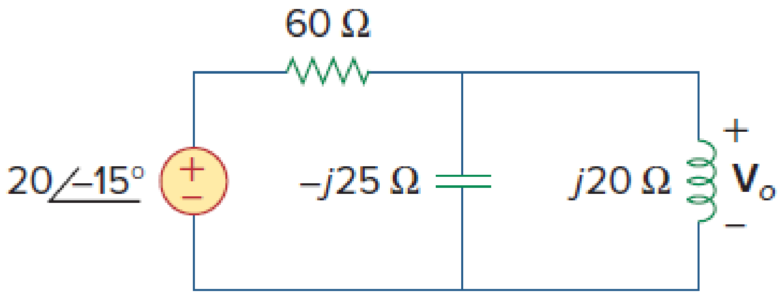
The voltage across a load is $\boldsymbol{v}(\boldsymbol{t}) = 1\mathbf{0}\sin(\boldsymbol{\omega t} + 30°)\boldsymbol{V}$ and the current through the element in the direction of the voltage drop is $\boldsymbol{i}(\boldsymbol{t}) = 2\sin(\boldsymbol{\omega t} + 60°)\boldsymbol{A}$ Find: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Find Z_{in}?



$$10\text{ mF} \quad \Rightarrow \quad \frac{1}{j\omega C} = \frac{1}{j4 \times 10 \times 10^{-3}} = -j25\ \Omega$$

$$5\text{ H} \quad \Rightarrow \quad j\omega L = j4 \times 5 = j20\ \Omega$$



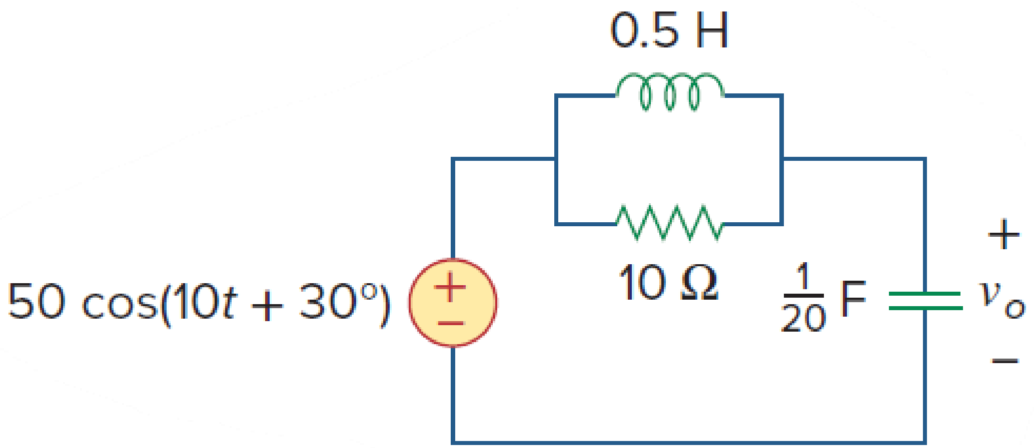
$\mathbf{Z}_1$ = Impedance of the 60-Ω resistor

$\mathbf{Z}_2$ = Impedance of the parallel combination of the 10-mF capacitor and the 5-H inductor

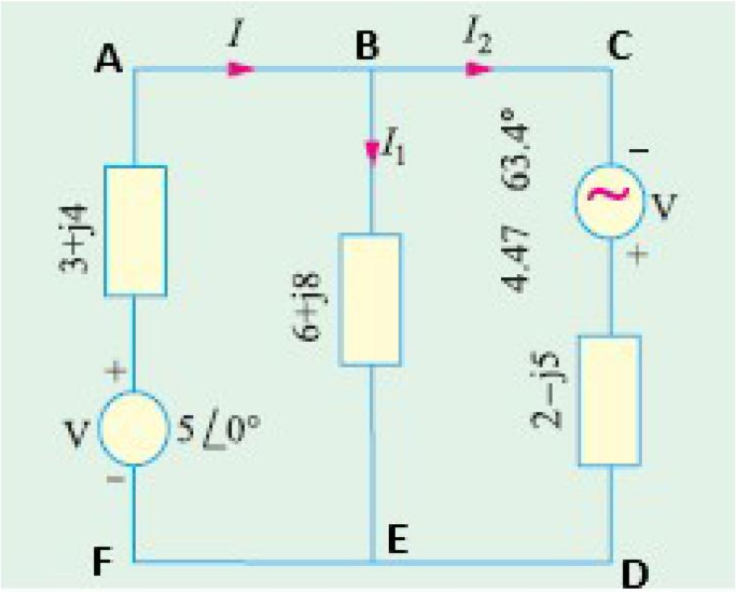
Then $\mathbf{Z}_1 = 60\ \Omega$ and

$$\mathbf{Z}_2 = -j25 \parallel j20 = \frac{-j25 \times j20}{-j25 + j20} = j100\ \Omega$$

Find Z_{in}?



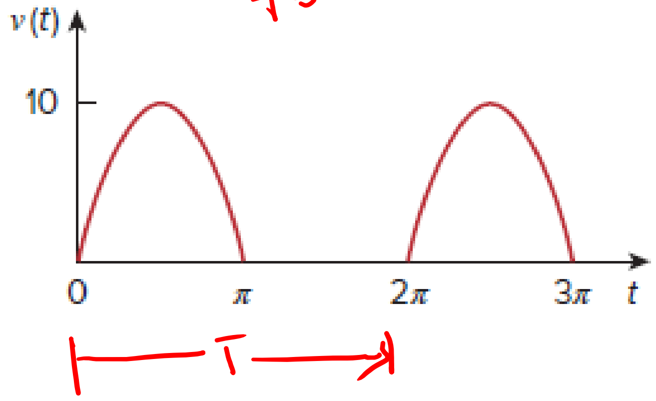
Using Kirchhoff's Laws, calculate the current flowing through each branch of the circuit shown in figure below.



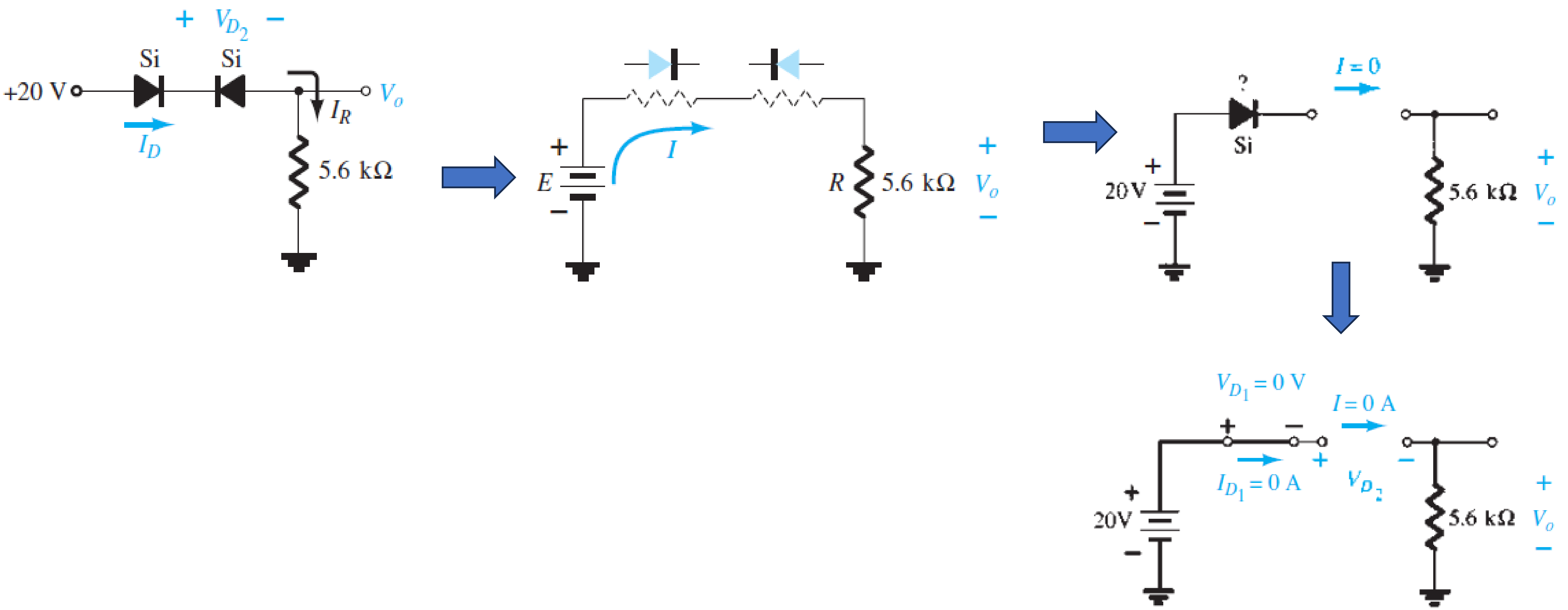
The waveform shown in Fig. below is a half-wave rectified sine wave. Find the rms value and the amount of average power dissipated in a 10-Ω resistor.

$v(t) = 10\sin\omega t$

$T = 2\pi$



Determine I_D , V_{D2}, and V_o for the circuit of Fig. below.



Determine I , V_1 , V_2 , and V_o for the series dc configuration of Fig below.